

**BANARAS HINDU UNIVERSITY**

B.A./B.Sc. (Hons) — Semester V Examination, 2022-23

**Paper No. MTB-501: Mathematical Analysis**

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

**Question 1.** (Compulsory.)

(a) Show that the limit of a convergent sequence is unique. [2]

(b) Show that the sequence  $(x_n)_{n=1}^{\infty}$ , where  $x_n = \left(1 + \frac{1}{n}\right)^n$ , is monotonically increasing. [2](c) Determine the values of  $x$  for which the series  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$  is absolutely convergent. [2](d) Test the convergence of the series  $\sum_{n=1}^{\infty} \frac{\cos(1/n)}{n^2}$ . [2](e) Let  $f(x)$  be defined by:

$$f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

Prove that  $f$  is Riemann integrable on  $\mathbb{R}$  (on any closed bounded interval). [2](f) Let  $f(x) = x$  on  $[0, 1]$ . If  $P = \left\{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\right\}$  is a partition of  $[0, 1]$ , then find:(i)  $L(P, f)$ , (ii)  $U(P, f)$ , (iii)  $\int_0^1 f dx$ , (iv)  $\overline{\int}_0^1 f dx$ . [4](g) If  $f(x)$  and  $g(x)$  are Riemann integrable on  $[a, b]$  such that  $f(x) \leq g(x)$  for all  $x \in [a, b]$ , show that  $\int_a^b f dx \leq \int_a^b g dx$ . [2](h) Test the convergence of  $\int_0^1 \frac{x}{1-x} dx$ . [2](i) Test the convergence of  $\int_0^{\infty} e^{-x^2} dx$ . [2](j) Show that  $\int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$ . [2]**Question 2.**

(a) Prove that the necessary and sufficient condition for the convergence of a monotonic sequence of real numbers is that it is bounded. [6]

(b) Prove that if a sequence  $(x_n)_{n=1}^{\infty}$  of positive numbers converges to a limit  $l$ , then:

$$\lim_{n \rightarrow \infty} (x_1 x_2 \cdots x_n)^{1/n} = l$$

Deduce that  $\lim_{n \rightarrow \infty} n^{1/n} = 1$ . [6]

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**Question 3.**

- (a) State and prove Cauchy's general principle for the convergence of sequences of real numbers. [6]
- (b) Let  $\sum_{n=1}^{\infty} a_n$  be a positive term series and let  $\lim_{n \rightarrow \infty} (a_n)^{1/n} = l$ . Then the series is:
- (i) convergent if  $l < 1$ ;
  - (ii) divergent if  $l > 1$ ;
  - (iii) the test fails if  $l = 1$ .

Prove this result (Cauchy's root test). [6]

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**Question 4.**

- (a) Test the convergence of the series  $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ . [6]
- (b) Define absolute and conditional convergence of a series. Show that the series  $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p}$  is absolutely convergent for  $p > 1$ , but only conditionally convergent for  $0 < p \leq 1$ . [6]
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**Question 5.**

- (a) Prove that a necessary and sufficient condition for the Riemann integrability of a bounded function  $f$  is that for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that for every partition  $P$  of  $[a, b]$  with norm  $\|P\| < \delta$ :

$$U(P, f) - L(P, f) < \varepsilon$$

[6]

- (b) Prove that every monotonic function on  $[a, b]$  is Riemann integrable on  $[a, b]$ . Give an example of a Riemann integrable function on  $[a, b]$  which is not monotonic on  $[a, b]$ . [6]
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**Question 6.**

- (a) Show that the function defined by:

$$f(x) = \frac{1}{n+1} \quad \text{for } \frac{1}{n+1} \leq x < \frac{1}{n}, \quad n = 1, 2, 3, \dots$$

is Riemann integrable on  $[0, 1]$ . Also show that  $\int_0^1 f(x) dx = \frac{\pi^2}{6} - 1$ . [6]

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- (b) If  $\int_a^b f(x) dx$  and  $\int_a^b g(x) dx$  both exist and  $g(x)$  keeps the same sign throughout the interval of integration, then there exists a number  $\mu$  lying between the bounds of  $f(x)$  such that:

$$\int_a^b f(x)g(x) dx = \mu \int_a^b g(x) dx$$

(State and prove the Second Mean Value Theorem for integrals.) [6]

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**Question 7.**

- (a) State and prove Dirichlet's test for improper integrals. [6]
- (b) State and prove the Limit Comparison Test (LCT) for improper integrals. Examine the convergence of  $\int_0^\infty \frac{\sin^p x}{x} dx$  for  $p > 1$ . [6]